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1 Overview

This AirView Exchange (AVX) technical specification only covers Detailed Data and Therapy Reporting APIs exposed by AVX, which are intended for use by integration partners.

Before you can use the APIs outlined in this document, you must first acquire the patient's EasyCare Number (ECN). Refer to the *AVX API Technical Specification for Patient Management* for instructions on how to get a patient's ECN.

The information in this technical specification supplements the *AVX API Technical Specification for Patient Management*. It aligns with Release 4.7 of AirView for the Americas and the Asia-Pacific region.

2 Uniform Resource Identifiers

The following table depicts the hierarchical structured design of AVX information. It also summarizes the HTTP calls that are available at each level.

URIs are specified in a manner that is consistent with RFC 6570 URI Template, and are case sensitive.

URI	Method	Description
https://<host>/v1/patients/{ecn}/reports/therapy?periodType={periodType}&startDate={startDate}&endDate={endDate}&noOfDays={days}&language={language}	GET	Returns a PDF therapy report for the requested ECN. See "3.1 Patient therapy report PDF" on page 4.
https://<host>/v1/patients/{ecn}/reports/complianceAndTherapy?periodType={periodType}&startDate={startDate}&endDate={endDate}&noOfDays={days}&language={language}	GET	Returns a PDF compliance and therapy report for the requested ECN. See "3.2 Patient compliance and therapy report PDF" on page 6.
https://<host>/v2/patients/{ecn}/reports/detailed?startDate={startDate}&endDate={endDate}&language={language}	GET	Returns a PDF detailed report for the requested ECN for a defined date range. See "3.3 Patient detailed report PDF" on page 8.
https://<host>/v1/patients/{ecn}/reports/detailed?noOfDays={days}&language={language}	GET	Returns a PDF detailed report for the requested ECN. See "3.3 Patient detailed report PDF" on page 8.

3 Detailed Data and Therapy Reporting Interfaces

3.1 Patient therapy report PDF

This API can be used to generate an AirView Therapy Report PDF for the specified patient and time period.

Property	Value
Method	GET
URL	<code>https://<host>/v1/patients/{ecn}/reports/therapy?periodType={periodType}&startDate={startDate}&endDate={endDate}&noOfDays={days}&language={language}</code>

where:

ecn is the patient unique identifier.

periodType is the type of report to return. See below for possible values.

startDate (optional) is the first date used to retrieve the patient therapy report for.

endDate (optional) is the last date to retrieve the patient therapy report for.

noOfDays (optional) is the number of consecutive days counting forward from the *startDate* or back from the *endDate* to generate a patient therapy report. The minimum number of days is one and the maximum number of days is 365.

Note: This date range includes the specified *startDate* or *endDate*.

language (optional) is the IETF BCP 47 code for the language to return the PDF report in.

Note: If the language parameter is omitted or unsupported, the report is returned in US English. See "4 Appendix A. Supported PDF languages" on page 10 for the list of supported languages.

Period types

Period type	Description
USERDEFINED	Returns the patient therapy report for the date range specified by the <i>startDate</i> , <i>endDate</i> and <i>noOfDays</i> parameters. Two of the following parameters are mandatory: <i>startDate</i> , <i>endDate</i> and <i>noOfDays</i> . If you use all three parameters, you will receive an error. Valid combinations are <i>startDate+endDate</i> , <i>startDate+noOfDays</i> and <i>endDate+noOfDays</i> .
COMPLIANCE	Returns the initial compliance report for the date range that AirView determined the patient to have met initial compliance. This includes the additional initial compliance section.

Note: When *periodType* is COMPLIANCE, the *startDate*, *endDate* and *noOfDays* parameters are ignored as the report returned is for the patient's initial compliance period.

Responses:

Success:

A PDF therapy report that meets the requested criteria and an HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/pdf
Body	n/a

Errors:

Property	Value
HTTP Response Status Code	Function of error – See table on the following page
Content Type	application/json
Body	JSON “Response” Object – See table on the following page

Field Name (JSON Name)	Type	Description
responseCode	string	Function of error – See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only
suggestedValues ()	object	Object containing valid parameters to use for the call
noOfDays	string	The number of consecutive days counting backwards from the endDate requested which have the same therapy mode
endDate	string	The endDate which was requested in the call

Description	Response Type	Error Code
Requesting user does not have access to the patient	HTTP 401	1002001
Therapy report requested but missing mandatory parameter	HTTP 400	1001002
Invalid parameter value	HTTP 400	1001003
Date not in correct format	HTTP 400	1001004
Therapy report requested but date range contains only unused days	HTTP 400	1014001
30-day compliance report requested but patient has not met compliance	HTTP 400	1014003
Therapy report requested but date range contains multiple therapy modes	HTTP 409	1014002

Example:

```
{
  "responseCode": "1014002",
  "suggestedValues": {
    "noOfDays": "2",
    "endDate": "2017-10-22"
  },
  "responseDesc": "Therapy Report date range contains multiple device modes. See suggestedValues"
}
```

Note: Identities and data are fictional.

3.2 Patient compliance and therapy report PDF

This API can be used to generate an AirView Compliance and Therapy Report PDF for the specified patient and time period.

Property	Value
Method	GET
URL	<code>https://<host>/v1/patients/{ecn}/reports/complianceAndTherapy? periodType= {periodType}&startDate={startDate}&endDate={endDate}&noOfDays={days}&language={language}</code> where: <i>ecn</i> is the patient unique identifier. <i>periodType</i> is the type of report to return. See below for possible values. <i>startDate</i> (optional) is the first date used to retrieve the compliance and therapy report for. <i>endDate</i> (optional) is the last date to retrieve the compliance and therapy report for. <i>noOfDays</i> (optional) is the number of consecutive days counting forward from the startDate or back from the endDate to generate a compliance and therapy report. The minimum number of days is one and the maximum number of days is 365. Note: This date range includes the specified startDate or endDate. <i>language</i> (optional) is the IETF BCP 47 code for the language to return the PDF report in. Note: If the language parameter is omitted or unsupported, the report is returned in US English. See "4 Appendix A. Supported PDF languages" on page 10 for the list of supported languages.

Period types

Period type	Description
USERDEFINED	Returns the compliance and therapy report for the date range specified by the startDate, endDate and noOfDays parameters. Two of the following parameters are mandatory: startDate, endDate and noOfDays. If you use all three parameters, you will receive an error. Valid combinations are startDate+endDate, startDate+noOfDays and endDate+noOfDays.
COMPLIANCE	Returns the initial compliance report for the date range that AirView determined the patient to have met initial compliance. This includes the additional initial compliance section.

Note: When periodType is COMPLIANCE, the startDate, endDate and noOfDays parameters are ignored as the report returned is for the patient's initial compliance period.

Responses:

Success:

A PDF Compliance and Therapy report that meets the requested criteria and an HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/pdf
Body	n/a

Errors:

Property	Value
HTTP Response Status Code	Function of error – See table below
Content Type	application/json
Body	JSON “Response” Object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	Function of error – See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only
suggestedValues ()	object	Object containing valid parameters to use for the call
noOfDays	string	The number of consecutive days counting backwards from the endDate requested which have the same therapy mode
endDate	string	The endDate which was requested in the call

Description	Response Type	Error Code
Requesting user does not have access to the patient	HTTP 401	1002001
Compliance and Therapy report requested but missing mandatory parameter	HTTP 400	1001002
Invalid parameter value	HTTP 400	1001003
Date not in correct format	HTTP 400	1001004
Compliance and Therapy report requested but date range contains only unused days	HTTP 400	1014001
30-day compliance report requested but patient has not met compliance	HTTP 400	1014003
Therapy report requested but date range contains multiple therapy modes	HTTP 409	1014002

Example:

```
{
  "responseCode": "1014002",
  "suggestedValues": {
    "noOfDays": "2",
    "endDate": "2017-10-22"
  },
  "responseDesc": "Therapy Report date range contains multiple device modes. See suggestedValues"
}
```

Note: Identities and data are fictional.

3.3 Patient detailed report PDF

This version 2 API can be used to generate an AirView Detailed Report PDF for the specified patient within a specified date range.

Property	Value
Method	GET
URL	https://<host>/v2/patients/{ecn}/reports/detailed?startDate={startDate}&endDate={endDate}&language={language} where: <i>ecn</i> is the patient unique identifier. <i>startDate</i> is the first date used to retrieve the detailed report for. <i>endDate</i> is the last date to retrieve the detailed report for. Note: This API attempts to return the inclusive number of days of known data between start and end dates excluding unused days. Zero usage days count as known data but are not included in the PDF. The minimum number of days in the date range is one and the maximum is 30. <i>language</i> (optional) is the IETF BCP 47 code for the language to return the PDF report in. Note: If the language parameter is omitted or unsupported, the report is returned in US English. See "4 Appendix A. Supported PDF languages" on page 10 for the list of supported languages.

Responses:

Success:

A PDF detailed report that meets the requested criteria and an HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/pdf
Body	n/a

Errors:

Property	Value
HTTP Response Status Code	Function of error – See table below
Content Type	application/json
Body	JSON "Response" Object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	Function of error – See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only

Description	Response Type	Error Code
Requesting user does not have access to the patient	HTTP 401	1002001
Detailed report requested but missing mandatory parameter	HTTP 400	1001002
Invalid parameter value	HTTP 400	1001003
Detailed report requested but date range contains only unused days	HTTP 400	1014001

This version 1 API can be used to generate an AirView Detailed Report PDF for the specified patient and number of days.

Property	Value
Method	GET
URL	https://<host>/v1/patients/{ecn}/reports/detailed?noOfDays={days}&language={language} where: <i>ecn</i> is the patient unique identifier. <i>noOfDays</i> is the number of days to generate the detailed report. This API attempts to return the last {noOfDays} days of known data excluding unused days. The last days of known data are not necessarily the last {noOfDays} calendar days. For example, if you generate a 30-day detailed report in March but the patient only has data from January 1-30, the API returns these dates. Zero usage days count as known data but are not included in the PDF. The maximum number of days is 30. <i>language</i> (optional) is the IETF BCP 47 code for the language to return the PDF report in. Note: If the language parameter is omitted or unsupported, the report is returned in US English. See "4 Appendix A. Supported PDF languages" on page 10 for the list of supported languages.

Responses:

Success:

A PDF detailed report that meets the requested criteria and an HTTP 200 response with the following fields:

Property	Value
HTTP Response Status Code	200
Content Type	application/pdf
Body	n/a

Errors:

Property	Value
HTTP Response Status Code	Function of error – See table below
Content Type	application/json
Body	JSON "Response" Object – See table below

Field Name (JSON Name)	Type	Description
responseCode	string	Function of error – See table below
responseDesc	string	Unspecified – Debugging purposes only
fieldName	string	Any applicable field name – Debugging purposes only

Description	Response Type	Error Code
Requesting user does not have access to the patient	HTTP 401	1002001
Detailed report requested but missing mandatory parameter	HTTP 400	1001002
Invalid parameter value	HTTP 400	1001003
Detailed report requested but date range contains only unused days	HTTP 400	1014001

4 Appendix A. Supported PDF languages

Locale ID	Report generated in this language
de	German
en	English ROW
en-us	English US
es	Spanish
fr	French
fr-ca	French-Canadian
ja	Japanese
pt	Portuguese
sv	Swedish
it	Italian



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